

D TULASI PRASAD

AI & ML Engineer | NLP & Deep Learning | Generative AI | Data Science
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PROFESSIONAL SUMMARY

Final-year B.Tech (CS & AI) graduate with hands-on experience building and deploying AI/ML systems. Developed a production-grade RAG pipeline using LangChain, FAISS, and OpenAI APIs, and deployed an NLP model via FastAPI and Docker achieving 89.5% accuracy on 50K+ samples. Proficient in Python, TensorFlow, PyTorch, Scikit-learn, and Pandas. Experienced in Generative AI, LLMs, and end-to-end ML workflows — from data preprocessing and model training to deployment and documentation. Strong analytical, problem-solving, and cross-team collaboration skills.

SKILLS

Languages	Python, SQL
ML / DL	Scikit-learn, TensorFlow, PyTorch, Keras - ANN, LSTM, GRU, CNN, Random Forest, SVM, PCA
NLP	TF-IDF, Word Embeddings, Transformers, Hugging Face, Text Classification
Generative AI	LangChain, LangGraph, LLMs, RAG, Prompt Engineering, Vector Databases, OpenAI API, MCP
MLOps / APIs	FastAPI, Docker, REST APIs, Git, GitHub
Data & Viz	Pandas, NumPy, Matplotlib, Seaborn, Streamlit, Power BI, MS Excel
Cloud	AWS (EC2, S3)

WORK EXPERIENCE

Data Science Intern (Lead Intern) | AI Variant

Jan 2026 – Apr 2026

- Led a team of interns on a World Development Measurement data science project - coordinating task allocation, code reviews, and delivery timelines across the full pipeline.
- Engineered and executed an end-to-end pipeline on the World Bank dataset: EDA across 200+ socioeconomic indicators, data cleaning, PCA-based dimensionality reduction, and K-Means clustering (best k=3, silhouette score 0.2965).
- Built an interactive 3D cluster visualization using Three.js and a Streamlit dashboard for stakeholder-facing presentation of country groupings and development patterns.

AI & ML Intern | SmartBridge Educational Services Pvt. Ltd.

Dec 2025 – Mar 2026

- Built and benchmarked ML models across 5+ classification tasks; improved baseline accuracy by ~12% via hyperparameter tuning (GridSearchCV) and L1/L2 regularization.
 - Delivered structured performance reports summarizing model trade-offs and actionable insights for stakeholders.
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PROJECTS

PDF RAG Q&A System — LangChain, ChromaDB, OpenAI, FastAPI, Streamlit, Docker [\[GitHub\]](#)

- Built a production-grade RAG pipeline: upload any PDF to ingest into ChromaDB vector store to answer natural language questions grounded in document context with source page attribution.
- Implemented PyPDFLoader and RecursiveCharacterTextSplitter chunking, OpenAI text-embedding-3-small embeddings, and a GPT-4o-mini LCEL chain for generation.
- Evaluated with RAGAS — achieved Faithfulness 1.0 (zero hallucination) and Answer Relevancy 0.91; deployed via FastAPI + Streamlit, fully containerized with Docker Compose.

Sentiment & Rating Analysis of Product Reviews — NLP, LSTM, FastAPI, Docker, Streamlit [\[GitHub\]](#)

- Performed EDA on 50,000+ Amazon reviews; engineered features using TF-IDF and sequence padding for 3-class sentiment classification.

- Built and benchmarked Logistic Regression vs. LSTM — best LSTM model achieved 89.5% accuracy (F1: 0.87), outperforming baseline by 8%.
- Deployed the full NLP pipeline as a FastAPI REST service with multiple endpoints, containerized with Docker for reproducible, production-ready delivery.

YouTube RAG Assistant — *LangChain, OpenAI, FAISS, Python* [\[GitHub\]](#)

- Built an end-to-end RAG pipeline ingesting YouTube transcripts and answering natural language questions grounded in video content using GPT-4o-mini and FAISS vector store.
- Implemented document chunking (1000-token chunks, 200-token overlap) and semantic embeddings with OpenAI text-embedding-3-small; assembled a production-style LangChain chain using RunnableParallel and RunnablePassthrough.

EDUCATION

B.Tech — Computer Science & Engineering (AI & Data Science) | Siddharth Institute of Engineering and Technology,
Tirupathi, AP 2022 – 2026
CGPA: 8.14 / 10.0

CERTIFICATIONS

Quantitative Research Job Simulation — J.P. Morgan Forage	<i>Mar 2026</i>
Data Science Certification — ExcelR Institute	<i>Mar 2026</i>
Data Science Internship certification – AI Variant	<i>Mar 2026</i>
Career Edge — Interview & Job Readiness + IT Primer — TCS iON	<i>Oct 2025</i>